

Working with Linked Lists

TIP101

Agenda

1 Welcome 🙌

2 Working with Linked Lists

3 **BREAK**

4 Collaborative Problem Solving

5 Wrap Up

10 minutes

35 minutes

5 minutes

60 minutes

10 minutes

Registration open now!

1:1 career sessions with tech professionals

Resume Review Week

July 20 - 23

Get personalized feedback on your resume before recruiting.

go.codepath.org/resume-reviews



Mentorship Week

July 27 - 30

Ask career questions + get advice.

go.codepath.org/mentorship

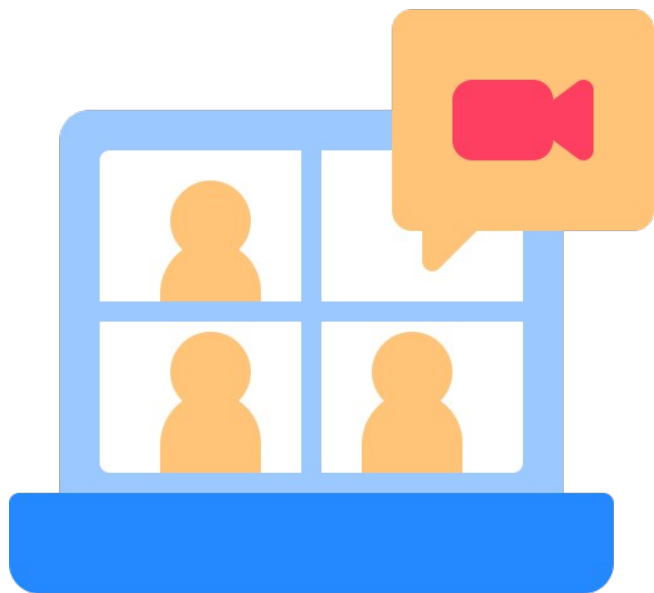


Don't wait to sign up - popular timeslots fill quickly!

CODEPATH*ORG

10 minutes

Welcome! 🖐️



**Turn on your
cameras**



**Fill out the entry
ticket**



Icebreaker

5 minutes



Question:

Who would you cast you play yourself in a movie about your life? No wrong answers!



Answer:

Share your answer in the Zoom chat below. What kinds of pets do you have if any? If you don't have any, share what type you would like to have, if any.

Your Feedback



Glows

- + First mock interview went well! Imagine with full time.
- + Gaining confidence as a class



Grows

- Will provide starter code to future volunteers
- Investigating adding test cases to the problem sets

35 minutes

Working with Linked Lists

Multiple Pass Technique

- * Traverse a linked list multiple times to extract information or process nodes.
- * Common applications:
 - Calculating the length of a list.
 - This is a required step for finding things in a “fixed” location (finding the middle node, second to last, etc).
 - Performing operations that depend on global information about the list.
 - “Count how many nodes have a value higher than the average value of all nodes.”

Temporary Head Technique

- * Create a temporary head node to handle edge cases involving the actual head.
- * When navigating through a linkedlist, we typically always want to be able to keep track of the head node. **This is typically what we will need to return.**
- * Common applications:
 - Deleting the head node without losing track of the rest of the list.
 - Adding a new node to the front of a list (updating the head).

Slow-Fast Pointer Technique

- * Use two pointers moving at different speeds to solve problems involving cycle detection or the middle element.
- * Common applications:
 - Detecting a cycle in a linked list.
 - Take a minute and think about how this would work.
 - Finding the middle node ($\frac{1}{2}$).
 - Determining the length of a cycle.



Instructor Demo

20 minutes

Write a function to find the middle node of a singly linked list. If the list has an even number of nodes, return the second middle node.

Examples:

* **Input:** head = [1, 2, 3, 4, 5]

Output: Node with value 3

* **Input:** head = [1, 2, 3, 4, 5, 6]

Output: Node with value 4



Instructor Demo

Naive Approach ("Brute Force")

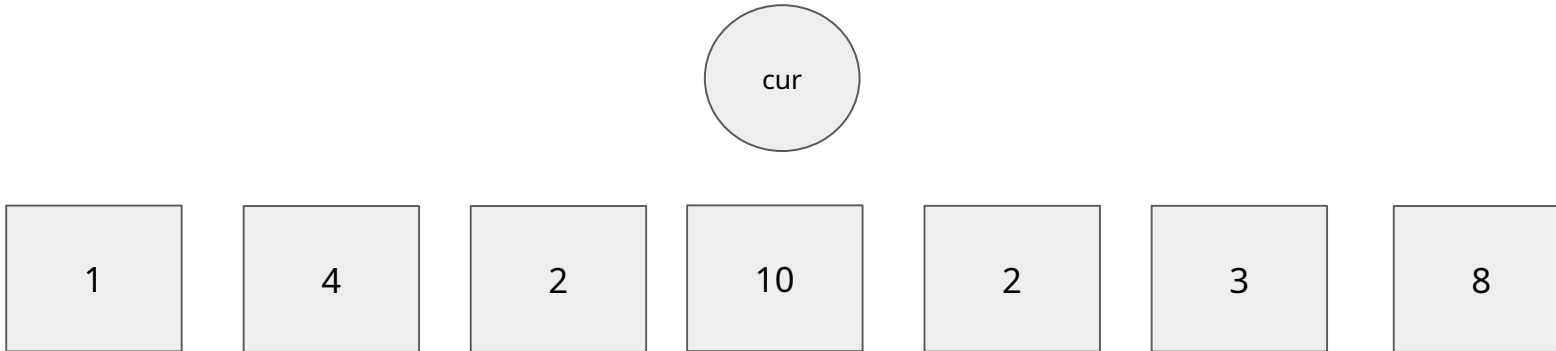
Multi pass

Counter variable, Cur, Head

count= 7

mid = $7//2$

END

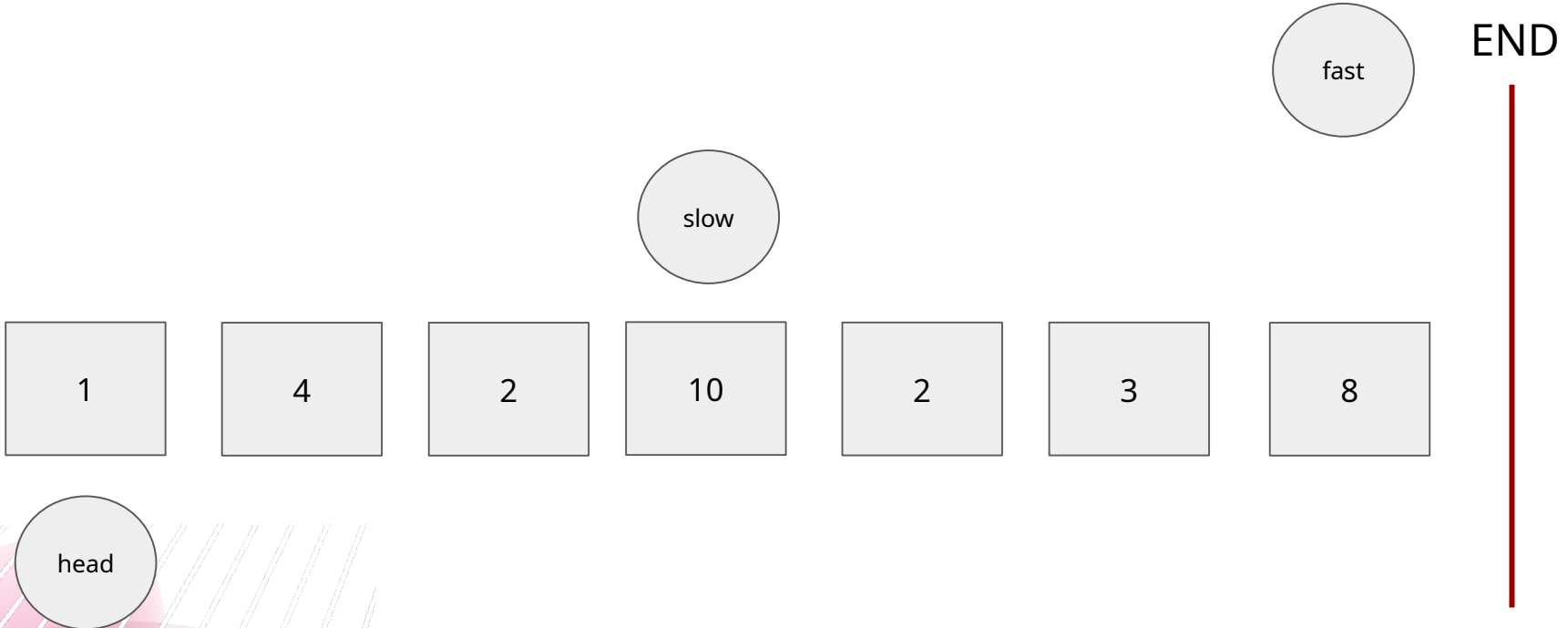




Instructor Demo

Optimized Approach

Slow, Fast, Head



Break Time!

Take 5 mins to step away from the computer. Feel free to turn off your camera and return promptly after 5 mins.



60 minutes

Collaborative Problem Solving



Breakout Rooms

60 minutes

1. 🏃 Navigate to your breakout room.
2. 🙌 Introduce yourselves!
3. 🧑💻 Work on the problems for your selected unit.

10 minutes

Wrap Up



One Good Thing

Let's reflect on today's session. Maybe you got a question answered, overcame a specific coding challenge, or learned how to do something completely new!

In the chat, share at least one good thing you've learned or experienced during the session today.

Let's celebrate together!



Session Survey Time!





Final Reminders

- ☐ Finish the problem sets for additional practice
- ☐ Take the weekly HackerRank assessment by Sunday @ midnight
- ☐ Visit a **Study Hall Session** to deepen your understanding, work through challenges, and connect with course material alongside your peers!
- ☐ Mock interview (Instructor + Volunteer) again on Thursday!